

Rosen — §9.1: Relations and Their Properties

Selected Exercises

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Exercise 8

Show that the relation $R = \emptyset$ on a nonempty set S is symmetric and transitive, but not reflexive.

Exercise 10

Give an example of a relation on a set that is

- (a) both symmetric and antisymmetric.
- (b) neither symmetric nor antisymmetric.

Exercise 15

Can a relation on a set be neither reflexive nor irreflexive?

Exercise 22

A relation R is called asymmetric if $(a, b) \in R$ implies that $(b, a) \notin R$.

Must an asymmetric relation also be antisymmetric? Must an antisymmetric relation be asymmetric? Give reasons for your answers.

Exercise 25

How many different relations are there from a set with m elements to a set with n elements?

Exercise 27

Let R be a relation from a set A to a set B . The inverse relation from B to A , denoted by R^{-1} , is the set of ordered pairs $\{(b, a) \mid (a, b) \in R\}$. The complementary relation $\overline{R}$ is the set of ordered pairs $\{(a, b) \mid (a, b) \notin R\}$.

Let R be the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set of positive integers. Find

- (a) R^{-1} .
- (b) $\overline{R}$.

Exercise 29

Suppose that the function f from A to B is a one-to-one correspondence. Let R be the relation that equals the graph of f . That is, $R = \{(a, f(a)) \mid a \in A\}$. What is the inverse relation R^{-1} ?

Exercise 33

Let R be the relation on the set of people consisting of pairs (a, b) , where a is a parent of b . Let S be the relation on the set of people consisting of pairs (a, b) , where a and b are siblings (brothers or sisters). What are $S \circ R$ and $R \circ S$?

Exercise 39

Let R be the relation on the set of people with doctorates such that $(a, b) \in R$ if and only if a was the thesis advisor of b . When is an ordered pair (a, b) in R^2 ? When is an ordered pair (a, b) in R^n , when n is a positive integer? (Assume that every person with a doctorate has a thesis advisor.)

Exercise 47

How many relations are there on a set with n elements that are

- (a) symmetric?
- (b) antisymmetric?
- (c) asymmetric?
- (d) irreflexive?
- (e) reflexive and symmetric?
- (f) neither reflexive nor irreflexive?

Exercise 48

How many transitive relations are there on a set with n elements if

- (a) $n = 1$?
- (b) $n = 2$?
- (c) $n = 3$?

Exercise 49

Find the error in the “proof” of the following “theorem.”

“Theorem”: Let R be a relation on a set A that is symmetric and transitive. Then R is reflexive.

“Proof”: Let $a \in A$. Take an element $b \in A$ such that $(a, b) \in R$. Because R is symmetric, we also have $(b, a) \in R$. Now using the transitive property, we can conclude that $(a, a) \in R$ because $(a, b) \in R$ and $(b, a) \in R$.

Exercise 52

Show that the relation R on a set A is antisymmetric if and only if $R \cap R^{-1}$ is a subset of the diagonal relation $\Delta = \{(a, a) \mid a \in A\}$.

Exercise 55

Let R be a relation that is reflexive and transitive. Prove that $R^n = R$ for all positive integers n .

Exercise 58

Let R be a symmetric relation. Show that R^n is symmetric for all positive integers n .

Exercise 59

Suppose that the relation R is irreflexive. Is R^2 necessarily irreflexive? Give a reason for your answer.